

Exam Choice

Student Number

2022 | TRIAL
EXAMINATION

Physics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black pen.
- Draw diagrams using pencil.
- For questions in Section II, show all relevant working in questions involving calculations.
- NESA approved calculators may be used.

Total marks: 100

Section I – 20 marks (pages 3 – 11)

- Attempt questions 1 – 20.
- Allow about 35 minutes for this section.

Section II – 80 marks (pages 12 – 26)

- Attempt questions 21 – 36
- Allow about 2 hours and 25 minutes for this section.

Section I – 20 marks

Attempt Questions 1-20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A  B  C  D 

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A  B  C  D 

correct 

1. A spacecraft with rest mass m accelerates using a force F , from a speed of $0.7c$ to $0.8c$. Afterwards, it uses the same force F to accelerate from $0.8c$ to $0.9c$.

The acceleration from $0.8c$ to $0.9c$, when compared with the acceleration from $0.7c$ to $0.8c$, will:

- (A) take the same time.
- (B) take more than twice the time.
- (C) take a somewhat longer time.
- (D) take a shorter time.

2. Projectile **A** is launched on Earth with an initial speed of 40 m s^{-1} at an angle of 30° to the horizontal.

Projectile **B** is launched with the same initial conditions on a planet where the acceleration due to gravity is exactly half that on Earth.

Projectile **B**'s path compared to projectile **A** will have:

	maximum height reached	final speed
(A)	equal	double
(B)	equal	equal
(C)	double	double
(D)	double	equal

3. If light had a pure wave nature, which observation from investigations into the photoelectric effect would occur?

- (A) There would be a delay in the initial emission of photoelectrons rather than immediate emissions observed.
- (B) The frequency of the incident light would still change the stopping voltage.
- (C) There would still be a threshold frequency.
- (D) The metal used in the cathode would have no effect on the photocurrent.

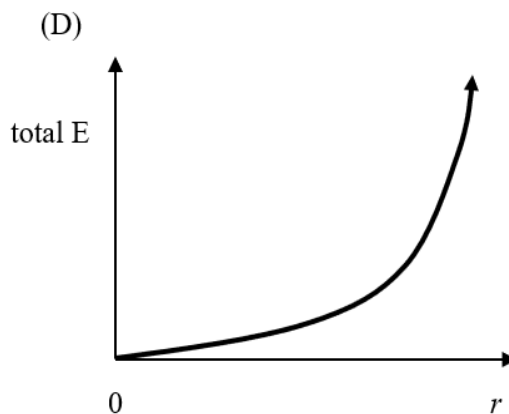
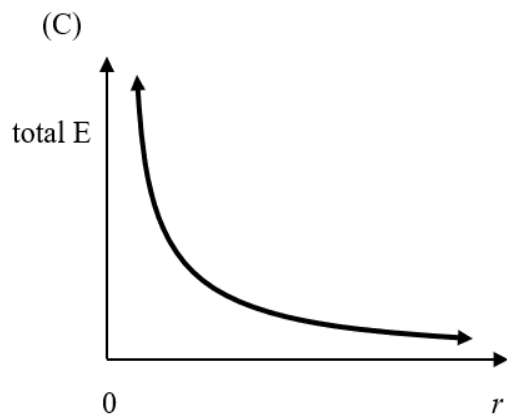
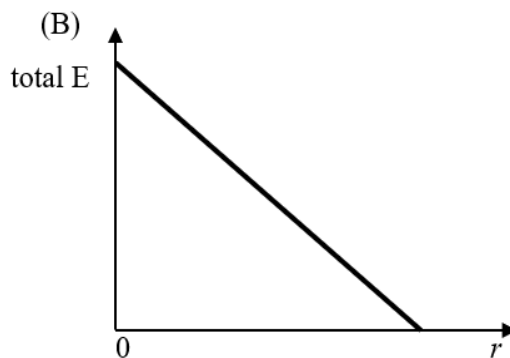
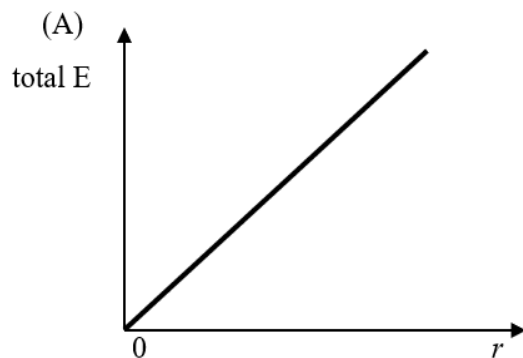
4. Two objects are launched into space from the ground so that they will just escape Earth's gravitational field forever, by being given an initial velocity.

Object **A** had a mass of 10 kg, while object **B** had a mass of 100 kg.

Which alternative is correct?

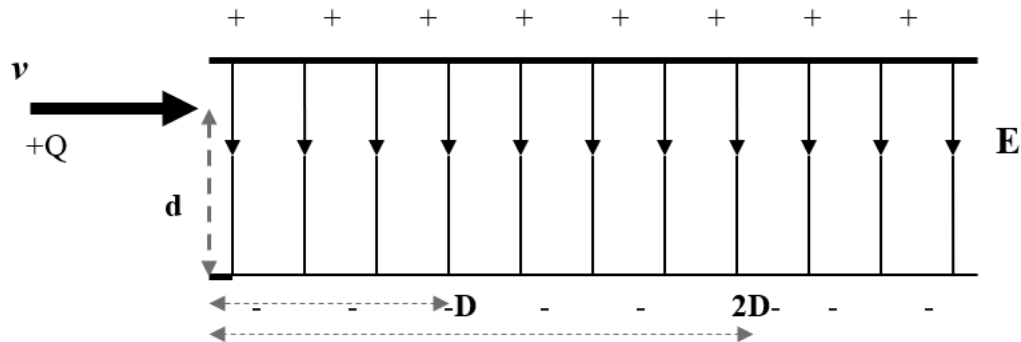
	Objects' initial speed	Objects' initial kinetic energy
(A)	equal	equal
(B)	equal	B is 10 times greater than A
(C)	equal	B is 100 times greater than A
(D)	B is 10 times greater than A	B is 100 times greater than A

5. Which graph best represents the change in a satellite's total energy as its orbital radius, r , changes?



6. A charged particle $+Q$ enters a region of uniform electric field E between two parallel plates, as shown. The particle's speed before entering is v .

The apparatus is in a vacuum and the experiment is being performed in weightless conditions. The charged particle travels a horizontal distance D before striking the charged plate.

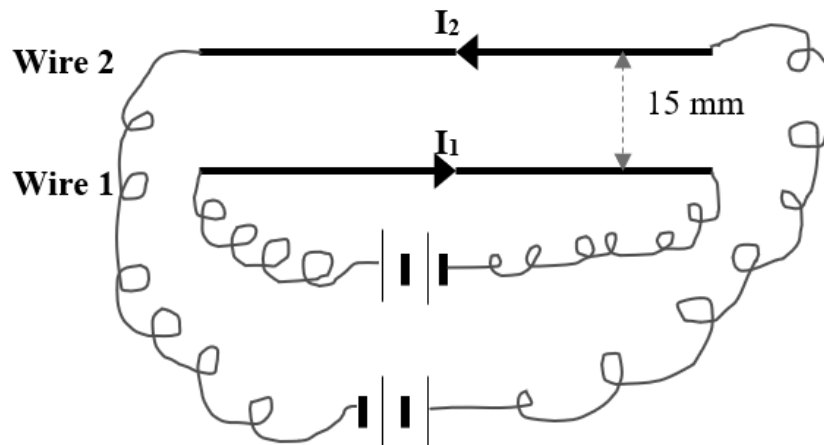


What change would result in the charged particle travelling twice the horizontal distance, i.e. $2D$?

- (A) doubling the initial velocity to $2v$
 - (B) halving the vertical distance, d , the particle must move
 - (C) halving the electric field strength
 - (D) halving the mass of the particle
7. An electron is accelerated between two charged plates, beginning from rest. As it moves between the plates, the electron's wavelength, λ ,
- (A) decreases
 - (B) stays constant
 - (C) increases
 - (D) remains zero

8. Two parallel wires, each of length l , are placed so that wire 2 is being suspended above wire 1, due to the force caused by the currents flowing in each wire, I_1 and I_2 balancing the weight of wire 2.

Wire 2 is suspended 15 mm above wire 1.



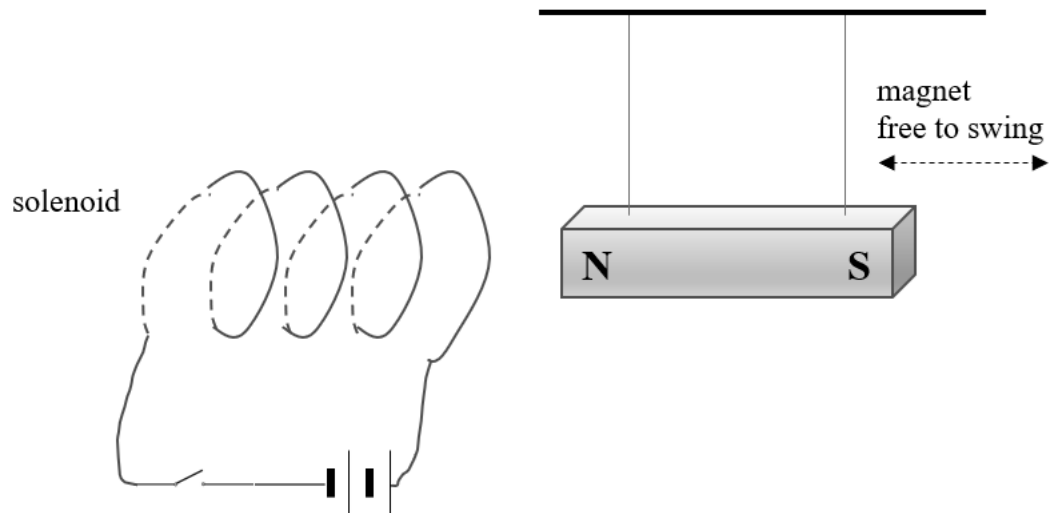
Both the currents, I_1 and I_2 are then doubled.

To what distance above wire 1 will wire 2 move to?

- (A) 22.5 mm
- (B) 30 mm
- (C) 45 mm
- (D) 60 mm

9. A solenoid is connected to a DC power source with a switch connected in series.

A magnet is placed near the end of the solenoid, suspended by cotton threads and free to swing.



At the instant that the switch is closed:

- (A) a back EMF would begin to be generated in the solenoid and the magnet would swing to the left.
 - (B) a surge of current would flow through the solenoid and the magnet would not move.
 - (C) a back EMF would be generated in the solenoid and the magnet would not move.
 - (D) a surge of current would flow through the solenoid and the magnet would swing to the right.
10. A step-down transformer has a primary voltage of 33 kV and a primary current of 200A. The secondary voltage is 11 kV with a secondary current of 460A.

The transformer's efficiency is closest to:

- (A) 90%
- (B) 75%
- (C) 50%
- (D) 33%

11. Maxwell's equations apply to the:

- (A) particle model of light only.
- (B) particle and wave models of light.
- (C) explanation of the photoelectric effect.
- (D) wave model of light only.

12. In an investigation designed to measure the speed of light, a single coherent, monochromatic beam of light with frequency 5.36×10^{14} Hz was split so that it travelled along two different paths before being re-combined and then observed using an interferometer.

When the path length difference was changed by 560 nm, what observations would be made about the re-combined light?

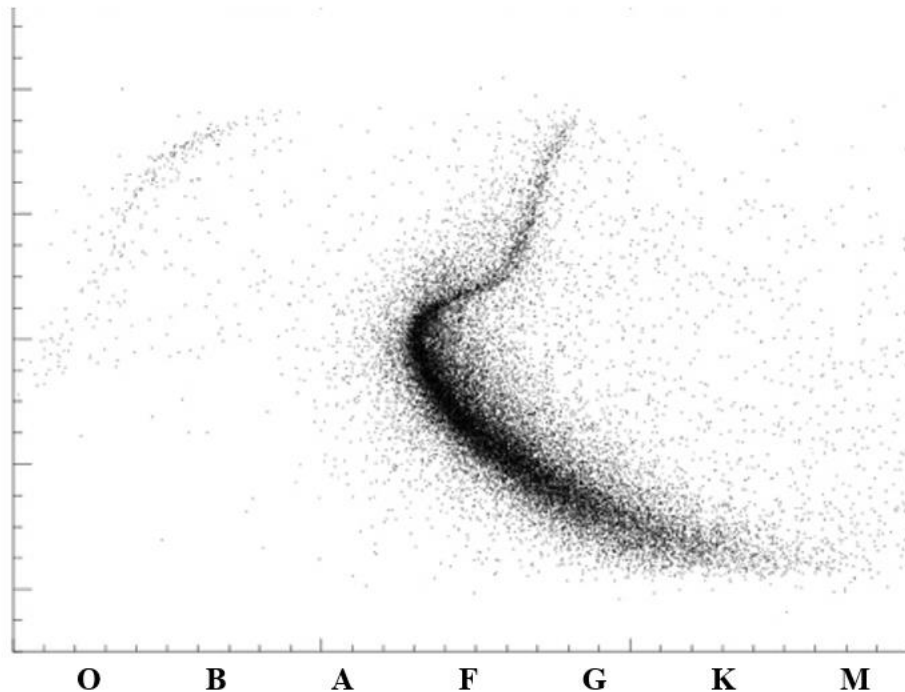
- (A) The intensity of the light would go through several cycles of constructive and destructive interference as the path length is being changed.
- (B) No change in the distribution of the intensity of the re-combined light would be observed compared to the original.
- (C) The intensity of the re-combined light after the change would be more than the original.
- (D) The intensity of the re-combined light after the change would be less than the original.

13. Laser light with a wavelength of 620 nm was passed through double slits with a separation of 60 μm . The screen was placed 2.00 m behind the slits.

How far from the centre of the central maximum is the next bright spot in the interference pattern on the screen?

- (A) 21 cm
- (B) 10 cm
- (C) 1.0 cm
- (D) 2.1 cm

14. The individual stars in a single star cluster are plotted on a Hertzsprung-Russell diagram. The results are shown below:



The relative under-representation of O and B type stars is most likely due to:

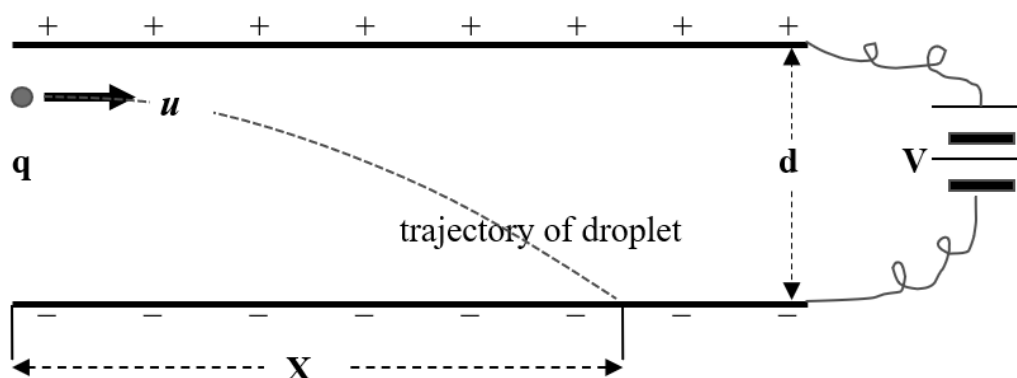
- (A) the cluster being too young for these stars to have formed yet.
 - (B) too much gas and dust existing between the stars preventing such stars from forming.
 - (C) these stars having depleted their hydrogen fuel and becoming red giants.
 - (D) the main sequence stars that formed in the cluster were not sufficiently large.
15. A fundamental particle is best described as being:

- (A) one that will not yield constituent particles in collisions in a particle accelerator.
- (B) one which has a partial charge.
- (C) a particle that makes up an atom.
- (D) protons, neutrons and electrons.

16. Which one of the statements below about nuclear fission is correct?
- (A) This is the process believed to provide the energy in stars.
 - (B) The sum of the masses of the particles produced is less than the sum of the mass of the original particle.
 - (C) The sum of the masses of the particles produced is greater than the sum of the mass of the original particle.
 - (D) Lighter particles collide and form a heavier particle, releasing energy.
17. A reason that the observations from the Geiger-Marsden experiment were unexpected at the time was that:
- (A) atoms were believed to have a positive nucleus.
 - (B) atoms were thought to be capable of reflecting alpha particles.
 - (C) electrons were believed to be orbiting the nucleus.
 - (D) atoms were thought to have the negative charges embedded throughout a continuous positive fluid.
18. The centripetal force required to maintain a 12.0 kg mass in horizontal circular motion with radius of 5.00m is 3.00×10^2 N.
- The speed of the object is:
- (A) 11.2 m s^{-1}
 - (B) 125 m s^{-1}
 - (C) 22.4 m s^{-1}
 - (D) 5.5 m s^{-1}

19. An investigation was performed to measure the charge on small oil droplets. The small oil droplets had an electric charge q applied to them before being fired with a known speed u into a region between two parallel charged plates, as shown.

The two plates have a known potential difference V between them and are separated by a known distance d .



The distance the droplets travelled before striking one of the charged plates, X , allowed for the calculation of the charge on the droplets.

Which other variables would need to have a known value for the calculation of the charge to be found?

- (A) The mass of the droplets and exactly where the droplets entered between the plates.
 - (B) The diameter of the oil droplets.
 - (C) The mass of the oil droplets and the size of the droplets.
 - (D) The strength of the electric field between the plates.
20. It is estimated that the Sun is producing 5.4×10^{25} J of energy each second by the conversion of mass to energy.

How many positron-electron annihilations would need to occur each second for an equivalent amount of energy to be produced?

- (A) 1.6×10^{19}
- (B) 6.0×10^8
- (C) 6.6×10^{38}
- (D) 3.3×10^{38}

2022

TRIAL
EXAMINATION

Physics

Section II Answer Booklet

80 marks

Attempt Questions 21 – 36

Allow about 2 hours and 25 minutes for this part

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in questions involving calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which questions you are answering.

Question 21 (5 marks)

Two projectiles are launched with the same initial speeds and angles to the horizontal. Projectile **A** has a mass of **m**, while projectile **B** has a mass of **2m**.

- (a) Explain why the two projectiles will have the same range. 2

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- (b) Both projectiles were launched again, this time with the same angles to the horizontal but with different initial speeds.

Projectile **B** remains in flight for twice as long as projectile **A** over level ground.

- How many times further does projectile **B** travel compared to projectile **A**? 3

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Question 22 (4 marks)

The train line between Sydney and Newcastle is designed so that it can take slow moving freight trains as well as much faster passenger trains at speeds up to 100 km h^{-1} . The track has many tight curves along the route.

- (a) Provide a reason why the track cannot be banked sufficiently so that a very fast train with speeds of 300 km h^{-1} could also use this track. 2

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- (b) For a section of track banked at 20° to the horizontal around a curve with a radius of 400 m, what is the design speed for a train? 2

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Question 23 (6 marks)

- (a) In the past, Earth's rotation was much faster than it is today.

Calculate the apparent weight of a 60.0 kg person standing on the equator if Earth's rotational period was 4 hours exactly.

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- (b) Identify **two** other factors which affect the gravitational field strength on the surface of the Earth and explain why they have an effect.

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Question 24 (5 marks)

Using the relationship between gravitational potential energy and kinetic energy, explain how the escape velocity from a distance r from the centre of mass of a

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planet is given by: $\sqrt{\frac{2GM}{r}}$ where M the mass of the planet.

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Question 25 (5 marks)

J. J. Thomson was able to find the charge to mass ratio of electrons by firing them into a region with a uniform magnetic field and observing their paths.

To do this, the electrons were first passed through a velocity selector, where electric and magnetic fields were aligned perpendicular to each other. Electrons with a particular speed were not deflected.

- (a) Explain why the two fields need to be perpendicular to each other. 2

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- (b) When using such a velocity selector with an electric field produced between two parallel plates 10.0 mm apart with a potential difference of 4.50×10^2 V, electrons with a speed of 6.50×10^4 m s⁻¹ passed through without being deflected.

Calculate the strength of the magnetic field within the velocity selector. 3

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Question 26 (5 marks)

Outline a safe procedure for an investigation to establish the relationship between the direction of the force on a straight current-carrying wire, the magnetic field and the current in the wire.

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Using specific cases not limited to those examples provided, explain how the interaction of electricity and magnetism and their respective fields is responsible for various phenomena studied in the Physics course.

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Marks

Question 28 (4 marks)

With the aid of diagrams, explain how a single current loop placed within a magnetic field can produce a torque for a full rotation, and thus act as a simple DC motor.

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Question 29 (7 marks)

Different types of spectra can be observed from different sources.

- (a) Outline the ways in which spectra from discharge tubes, reflected sunlight and incandescent filaments are produced. 4

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- (b) Identify which of these spectra can be used to identify elements, and explain how such an application is possible 3

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Question 30 (6 marks)

Observations of black body radiation could not be explained using a wave model of light.

- (a) Discuss how Planck's contribution to the explanation of radiation from a black body led to a change in the model of light. **4**

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- (b) Wien's Law can be used to find the wavelength of the peak of the radiated energy from a black body. However, for a body at 0 K (absolute zero), this is not possible. Why is this so? **2**

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Question 31 (4 marks)

The Hafele-Keating experiment involved flying extremely precise atomic clocks onboard aircraft in different directions around the world and comparing the times on the clocks when brought back together.

Discuss how the Hafele-Keating experiment was able to verify an aspect of special relativity as proposed by Einstein.

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Question 32 (3 marks)

Hubble's observations gave clear evidence that the Universe is expanding.

Describe what observations Hubble made, how they were made and outline the subsequent calculations that led to this conclusion.

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Question 33 (4 marks)

Given that human eyes can detect light with wavelengths from 400 – 700 nm, show, using appropriate calculations, that at least four emission lines in the Balmer series can be seen with the human eye.

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Question 34 (4 marks)

A radioactive sample of iodine-125 has a half-life of 60 days.

(a) What is the decay constant of iodine-125?

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(b) After 10 days, to what level of the original radioactivity has the sample fallen? Express your answer as a percentage of the original sample.

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Question 35 (6 marks)

- a) Outline the similarities and differences between alpha, beta and gamma decay.

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- (b) When 2 protons fuse to form a deuterium nucleus, a positron and a neutrino are released.

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Calculate the binding energy per deuterium nucleus formed, assuming the mass of a neutrino to be negligible.

Assume the mass of a deuterium nucleus to be 3.3435×10^{-27} kg.

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Question 36 (3 marks)

Particle accelerators make use of relativistic effects by providing particles with kinetic energies well in excess of those predicted using non-relativistic physics.

With reference to relevant quantitative physics, explain why particles accelerated in this way are essential for providing evidence for and testing theories of matter.

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If you use this space indicate clearly which question you are answering.

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Section II extra writing space.

If you use this space indicate clearly which question you are answering.

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